contractors are asked to include a study plan in their proposal with detailed plans and procedures for conducting the nutrient analysis of Key Foods, as well as identifying the analytical methods and procedures they will use to complete each task. The description of analytical methods includes sample handling and storage, extraction or digestion, analysis, and quantification steps performed as part of the analysis. The laboratories propose specific analytical methods, based on their expertise, which are examined by NDL during the review of the proposals. A detailed discussion of day-to-day quality control (QC) procedures is requested to facilitate the assessment of accuracy and precision for the unknown samples. The commercial laboratory proposals are evaluated by a panel consisting of NDL and other ARS staff members. The proposals are reviewed and scored against criteria delineated in the Request for Proposals.

Those offerors whose proposals are deemed technically acceptable are sent "check" samples by FALCC for analysis. These are Certified Reference Materials (CRMs) procured from a variety of sources, both in the US and at the global level. Nutrient-specific analytical results generated by offerors for these samples are evaluated against acceptable ranges prepared by NDL. Offerors with the best written proposals and analytical results on the check samples may be awarded a contract for specific nutrients. Specific work orders under each contract are awarded such that contractors will not be given analytical work for nutrients where results for the check samples were outside the acceptable range.

Aliquots of each food composite are sent to the laboratories by FALCC for analysis according to the work plans developed by NDL. The methods of analysis employed by the various analytical laboratories are given in Table 20. Along with the samples, FALCC includes a QC material, which is either a control composite developed at FALCC or a CRM purchased from a certifying organization (Phillips *et al*., 2006). The laboratories are required to provide the results of their in-house quality control runs with the results for the analytical samples. The results from the laboratories are then reviewed by a quality control committee comprised of NDL and FALCC staff. The QC data for CRMs are compared to the certificate values for the material and the results for control composites are compared to a database of all results obtained for the particular control composites. Analytical data for food samples are compared to existing data for that food or a similar food. Questions are referred to the laboratories, and, if necessary, the analyses are repeated.

Table 20. Methods of analysis used by NFNAP laboratories

Nutrient	Technique	Methods Identification
Protein (Nitrogen)	Combustion	AOAC 968.06 (4.2.04) Protein (Crude) in Animal Feed
	Combustion	AOAC 990.03 Protein (Crude) in Animal Feed
	Combustion	AOAC 992.15 (39.1.16) Crude Protein in Meat and Meat Products Including Pet Foods
	Kjeldahl	AOAC 991.20 Nitrogen (Total) in Milk
Total Fat	Acid hydrolysis	AOAC 989.05 (33.2.26) Fat in Milk, Mojo, Acid Hydrolysis
	Acid hydrolysis	AOAC 922.06 (32.1.14) Fat in Flour, Acid Hydrolysis Method
	Acid hydrolysis	AOAC 925.12 (32.5.05) Fat in Macaroni Products

Nutrient	Technique	Methods Identification
	Acid hydrolysis	AOAC 954.02 (4.5.02 or 7.063) Fat (Crude) or Ether Extract in Pet Food
	Extraction	AOAC 920.39 Fat (Crude) or Ether Extract in Animal Feed
	Extraction	AOAC 933.05 Fat in Cheese
	Extraction	AOAC 960.39 (39.1.05) Fat (Crude) or Ether Extract in Meat
	Extraction	AOAC 983.23 (45.4.02) Fat in Foods, Chloroform-Methanol Extraction Method
	Extraction	Folch *et al*., (1957) J. Biol. Chem., 226; 497-509.
	Extraction	Phillips *et al*. Simplified Gravimetric Determination of Total Fat in Mixed Food Composites After Chloroform/Methanol Extraction J. Amer. Oil Chem. Soc., 74 (1997)p. 137-142
	Extraction	AOAC 989.05 Fat in Milk
Ash	Gravimetric	AOAC 923.03 (32.1.05 or 14.006) Ash of Flour
	Gravimetric	AOAC 942.05 (4.1.10) Ash of Animal Feed
	Gravimetric	AOAC 945.46 Ash of Milk
Moisture	Vacuum oven	AOAC 934.01 (4.1.03) Moisture in Animal Feed
	Vacuum oven	AOAC 934.06 (37.1.10) Moisture in Fruits, Vegetables, and their Products
	Vacuum oven	AOAC 964.22 (42.1.05) Solids (Total) in Canned Vegetables: Gravimetric Method
	Forced air	AOAC 950.46 (39.1.02) Moisture in Meat
Fiber	Enzymatic-gravimetric	AOAC 991.43 (32.1.17) Total, Soluble, and Insoluble Dietary Fiber in Foods
	Enzymatic-gravimetric	AOAC 985.29 (45.4.07) Total Dietary Fiber in Foods
Starch	Enzymatic-colorimetric	AOAC 979.10 (32.2.05) Starch in Cereals, Glucoamylase Method
	Polarimetric	The Feedings Stuffs (Sampling and Analysis) Regulations 1982 No. 1144, Agriculture, London
Sugars	LC	AOAC 982.14 (32.2.07) Glucose, Fructose, Sucrose, and Maltose in Presweetened Cereals
Minerals	ICP	AOAC 984.27 Ca, Cu, Fe, Mg, Mn, P, K, Na and Zn in Infant Formula
	Atomic absorption	Laboratory modified AOAC 968.08 (4.8.02) + 985.35 (50.1.14) + 965.05 (2.6.01) Metals in Food by AAS
	ICP	Laboratory modified AOAC 985.01 (3.2.06) + 984.27 (50.1.15) Metals in Food by ICP
Selenium	Isotope dilution GC/MS	Reamer & Veillon, Anal. Chem., 53, (1981) 2166
	Hydride generation	AOAC 986.15 (9.1.01) Arsenic, Cadmium, Lead, Selenium and Zinc in Human and Pet Foods
Retinol	HPLC	AOAC 974.29 (modified for HPLC) Vitamin A in Mixed Feeds, Premixes, and Foods and Int'l Vitamin Nutrition (1992) (modified for HPLC determination) or a laboratory modified method with UV & fluorescent detection
Fluoride	Specific ion electrode	VanWinkle, Levy *et al*., Pediatr. Dent., 17 (1995) p305 (direct-read)
	Microdiffusion	VanWinkle, Levy *et al*., Pediatr. Dent., 17 (1995) p305 (microdiffusion)
Vitamin E	GC	Cort *et al*., J Agr Food Chem (1983) 31:1330-1333 + Speek *et al*., J Food Sci (1985) 50:121-124 + McMurray *et al*., J AOAC (1980) 63:1258-1261

Nutrient	Technique	Methods Identification
	LC	Ye, Landen, Eitenmiller J Agric Food Chem. 2000 Sep;48(9):4003-8.
Carotenoids	HPLC	AOAC 941.15 (45.1.03) modified by Quackenbush, J. Liq. Chroma. (1987) 10:643-653
	HPLC	Craft, N. 2001. Chromatographic techniques for carotenoid separation. *In* Current Protocols in Food Analytical Chemistry. F2.3.1−F2.3.15. Wrolstad, R. E., Acree, T. E., Decker, E. A., Penner, M. H., Reid, D. S., Schwartz, S. J., Shoemaker, C. F., Sporns, P., Editors. Wiley.. New York.
Thiamin	Fluorometric	AOAC 942.23 Thiamine (B1) in Foods
Riboflavin	Microbiological	Laboratory modified AOAC 940.33 (45.2.06) Riboflavin (Vitamin B2) in Vitamin Preparations
	Fluorometric	AOAC 970.65 Riboflavin (Vitamin B2) in Foods and Vitamin Preparations (Fluorometric)
Niacin	Microbiological	Laboratory modified AOAC 944.13 (45.2.04) Niacin and Niacinamide (Nicotinic Acid and Nicotinamide) in Vitamin Preparations
Pantothenic acid	Microbiological	AOAC 945.74 (45.2.05) Pantothenic Acid in Vitamin Preparations
	Microbiological	AOAC 992.07 (50.1.22) Pantothenic Acid in Milk-Based Infant Formula
Vitamin B6	Microbiological	AOAC 961.15 (45.2.08) Vitamin B6 (Pyridoxine, Pyridoxal, and Pyridoxamine) in Food Extracts (Microbiological)
Vitamin B12	Microbiological	AOAC 952.20 (45.2.02) Cobalamin (Vitamin B12 Activity) in Vitamin Preparations
Total folate	Microbiological	Martin *et al*. J Assoc Off Anal Chem. 1990 Sep-Oct;73(5):805-8.
Choline	LC/ESI/MS	Koc *et al*. (Zeisel), Quantitation of Choline and its Metabolites in Tissues and Foods by LC/ESI/MS. Anal. Chem. (2002) 74:4734-4740
Vitamin D	LC	AOAC 995.05 (50.1.23) Vitamin D in Infant Formulas and Enteral Products
	HPLC	AOAC 982.29 (45.1.22) Vitamin D in Mixed Feeds, Premixes, and Pet Foods
	HPLC	Birdwell *et al*. Am J Clin Nutr **88** (2008) 554S-557S
	LC/MS/MS	Huang, Luzerne, Winters & Sullivan, JAOAC Int., 92 (2009) p1327-1335
Vitamin C	Microfluorometric	AOAC 967.22 (45.1.15) Vitamin C (Total) in Vitamin Preparations
Vitamin K	HPLC	Booth & Sadowski, Methods Enzymol., (1997) 282:446 (HPLC)
Cholesterol	GC/Direct saponification	AOAC 994.10 (45.4.10) Cholesterol in Foods
	GC/Direct saponification	Dinh *et al*. J Food Comp Anal, 21 (2008) p306-314
	Acid Hydrolysis-HPLC	AOAC 982.30 (45.3.05) (modified) Protein Efficiency Ratio (Ninhydrin post column)
	Alk. hydrolysis-HPLC	AOAC 988.15 (modified) Tryptophan in Foods and Food and Feed Ingredients
	Colorimetric	990.26 (39.1.27) Hydroxyproline in Meat and Meat products
	Performic oxidation-HPLC	994.12 (4.1.11) (modified) Amino Acids in Feed (OPA post column)

Nutrient	Technique	Methods Identification
Amino acids	Alk. hydrolysis-HPLC	AOAC 988.15 (modified) Tryptophan in Foods and Food and Feed Ingredients
	Performic oxidation-HPLC	AOAC 994.12 (4.1.11) (modified) Amino Acids in Feed (OPA post column)
	Acid Hydrolysis-HPLC	AOAC 982.30 (45.3.05) (modified) Protein Efficiency Ratio (Ninhydrin post column)
	Colorimetric	AOAC 990.26 (39.1.27) Hydroxyproline in Meat and Meat products
Fatty acids	GLC	CE 1-62 (1997) Fatty Acid Composition by Gas Chromatography
	GLC	AOCS Ce 1-62 for GC, and Ce 2-66 for prep of methyl esters
	GLC	AOAC 996.06 (41.1.28A) Fat (Total, Saturated and Monounsaturated) in Foods
	GLC	AOAC 996.06 (41.1.28A) Fat (Total, Saturated, and Unsaturated) in Foods & AOCS Ce 1c-89 Fatty Acid Composition by Gas Chromatography (modified)

Compile newly generated data to update the National Nutrient Databank

The acceptable data from the analytical laboratories are then combined with the descriptive information collected on the sample units and are migrated to NDL's Nutrient Databank System, which was designed with three levels (Initial, Aggregated, and Compiled) to manage and process food composition data (Haytowitz *et al*, 2009). In the Initial step, all the individual data points are maintained along with complete information on methods of analysis, analytical quality control, sample handling, common measures, component and refuse data, and the source and sampling information for each sample unit. Information is also retained on how individual sample units are composited. Values are converted to standard units of measure per 100 g, but the "as received" data values are also retained. In the Aggregated step, NDL scientists make decisions on how to group the data (e.g., combining data from different sources or a single source), weight the data (usually by market share or production information), and/or handle new data when data already exist for a food item (i.e., replace the old data or combine it with new data). Specialized statistical procedures are used to aggregate the groups of data and generate descriptive statistics which take into consideration the grouping and nature of the data. NDL scientists also use statistical procedures within the NDBS to compare sets of data and test for outliers. For food items used in the FNDDS, missing data are imputed according to scientific principles (Schakel *et al.,* 1998) at the Compiled step. Missing values can be calculated using the recipe or formulation modules within the databank system. These modules are based on linear regression and are often used to generate a few missing values for some foods and complete nutrient profiles for other foods. The formulation regression program uses nutrient values and ingredients (in a specified order) from product labels. The recipe program uses known amounts from authoritative sources to generate a specific food nutrient profile. Finally, the name of the food item is finalized, common measures are selected and ranked, and the item is approved for release. Prior to release, the data are sent to experts for review; brand name items are sent to food companies or appropriate trade associations, and other

foods are sent to analysts or other specialists familiar with the food and its nutrient content. The experts indicate if the data are acceptable based on their knowledge of the products and if any changes are recommended. If changes are made, the data are disseminated in annual releases of the SR database.

References for Notes on Foods – National Food and Nutrient Analysis Program

Dietary Guidelines Advisory Committee (DGAC). *Report of the Dietary Guidelines Advisory Committee on the Dietary Guidelines for Americans, 2010***.** Version current 2010. Internet: http://www.cnpp.usda.gov/DGAs2010-DGACReport.htm (accessed 30 June 2014).

Haytowitz DB, Pehrsson PR, Holden JM. Setting priorities for nutrient analysis in diverse populations. J Food Comp Anal 2000;13:425-33.

Haytowitz DB, Pehrsson PR, Holden JM. The identification of key foods for food composition research. J Food Comp Anal 2002;15:183-94.

Haytowitz DB, Lemar LE, Pehrsson PR. USDA's nutrient databank system—a tool for handling data from diverse sources. J Food Comp Anal 2009;22:433-441.

Holden JM, Bhagwat SA, Patterson KY. Development of a multi-nutrient data quality evaluation system. J Food Comp Anal 2002;15:339-48.

Holden JM, Bhagwat SA, Haytowitz D, Gebhardt S, Dwyer J, Peterson J, Beecher GR, Eldridge AL. Development of a database of critically evaluated flavonoid data: application of USDA's data quality evaluation system. J Food Comp Anal 2005;18:829-44.

National Center for Health Statistics (NCHS). Center for Disease Control and Prevention, US Department of Health and Human Services. National Health and Nutrition Examination Survey 2007-2008. Version current 2010. Internet: http://wwwn.cdc.gov/nchs/nhanes/search/nhanes07_08.aspx (accessed 30 June 2014).

Pehrsson PR, Haytowitz DB, Holden JM, Perry CR, Beckler DG. USDA's national food and nutrient analysis program: food sampling. J Food Comp Anal 2000;12: 379-89.

Pehrsson PR, Perry CR, Cutrufelli RL, Patterson KY, Wilger J, Haytowitz DB, Holden JM, Day CD, Himes JH, Harnack L, et al. Sampling and initial findings for a national study of fluoride in drinking water. J Food Comp Anal 2006;19:S45-52.

Perry CR, Beckler DG, Bellow ME, Gregory LG, Pehrsson PR. Alaska Native and American Indian Tribe Sampling Frame Construction and Sample Design for the National Food and Nutrient Analysis Program. Version current 2001. Internet: http://www.amstat.org/sections/srms/proceedings/y2001f.html (accessed 30 June 2014).

Perry CR, Pehrsson PR, Holden J. A Revised Sampling Plan for Obtaining Food Products for Nutrient Analysis for the USDA National Nutrient Database. Version current 2003. Internet: http://www.amstat.org/sections/srms/proceedings/y2003f.html (accessed 30 June 2014).

Perry, CR, Daniel, MG, and Pehrsson, PR. A Revised Sampling Plan for Obtaining Food Products for Nutrient Analysis under the National Food and Nutrient Analysis Program: Selection of Sample. In *JSM Proceedings*, Government Statistics Section. Montreal, Canada: American Statistical Association 2013; 3884-3898.

Phillips KM, Wunderlich KM, Holden JM, Exler J, Gebhardt SE, Haytowitz D, Beecher GR, Doherty RF. Stability of 5-methyltetrahydrofolate in frozen fresh fruits and vegetables. Food Chem 2005;92:587-95.

Phillips KM, Patterson KY, Rasor AS, Exler J, Haytowitz DB, Holden JM, Pehrsson, PR. Quality-control materials in the USDA national food and nutrient analysis program (NFNAP). Anal Bioanal Chem 2006;384(6):1341-55.

Schakel SF, Buzzard IM, Gebhardt SE. Procedures for estimating nutrient values for food composition databases. J Food Comp Anal 1998;10:102-14.

US Census Bureau. US Census Bureau Population Estimates 2001. Version current 2002. Internet: http://www.census.gov/popest/data/historical/2000s/vintage_2002/index.html (accessed 14 July 2014).

US Department of Agriculture, Agricultural Research Service, Food Surveys Research Group. USDA Food and Nutrient Database for Dietary Studies, Release 5.0 [online database]. Version current 2012. Internet: http://www.ars.usda.gov/Services/docs.htm?docid=12089 (accessed 30 June 2014).

US Department of Agriculture, Agriculture Research Service, Nutrient Data Laboratory. USDA Database on the Choline Content of Common Foods - 2004. Version current 2004. Internet: http://www.ars.usda.gov/Services/docs.htm?docid=6232 (accessed 30 June 2014).

US Department of Agriculture, Agriculture Research Service, Nutrient Data Laboratory. USDA Database for the Proanthocyanidin Content of Foods – 2004. Version current 2004. Internet: http://www.ars.usda.gov/Services/docs.htm?docid=5843 (accessed 30 June 2014).

US Department of Agriculture, Agriculture Research Service, Nutrient Data Laboratory. USDA National Fluoride Database of Selected Beverages and Foods, Release 2.0. Version current 2005. Internet: http://www.ars.usda.gov/Services/docs.htm?docid=6312 (accessed 30 June 2014).

US Department of Agriculture, Agriculture Research Service, Nutrient Data Laboratory. National Nutrient Database for Standard Reference Release 26. Version current 2012. Internet: http://www.ars.usda.gov/Services/docs.htm?docid=8964 (accessed 30 June 2014).

US Department of Agriculture, Agriculture Research Service, Nutrient Data Laboratory. USDA Database on the Flavonoid Content of Foods Release 3.0. Version current 2013. Internet: http://www.ars.usda.gov/Services/docs.htm?docid=6231 (accessed 30 June 2014)

US Department of Agriculture, Agriculture Research Service, Nutrient Data Laboratory. USDA for the Isoflavone Content of Selected Foods Release 2.0. Version current 2008. Internet: http://www.ars.usda.gov/Services/docs.htm?docid=6382 (accessed 30 June 2014).

US Department of Agriculture, Agriculture Research Service, Nutrient Data Laboratory. National Nutrient Database for Standard Reference Release 23. Version current 2010. Internet: http://www.ars.usda.gov/Services/docs.htm?docid=22115 (accessed 30 June 2014).

US Department of Agriculture (USDA) and US Department of Health and Human Services (USDHHS). Dietary Guidelines for Americans, 2010. 7th ed. Version current 2011. Internet: http://www.cnpp.usda.gov/DGAs2010-PolicyDocument.htm (accessed 30 June 2014).

Beef Products (Food Group 13)

Introduction

Data for beef products are presented in the USDA National Nutrient Database for Standard Reference. For most retail cuts, nutrient values are presented for cuts trimmed to 1/8-inch and 0-inch fat and for Choice or Select quality grades. Nutrient values reported as "All Grades" were estimated by combining the nutrient values for Choice and Select grades, weighted by their market proportions. A few Prime cuts trimmed at 1/8 inch external fat are also included.

The data in SR represent the amount of each constituent in 100 grams of edible portion. The edible portion in beef may be represented as "separable lean and fat" or as "separable lean only". In both cases, bone and connective tissue are removed from the cut and reported as refuse. In the case of "separable lean and fat", it is assumed that all fat present is consumed. For items described as "separable lean only", all external trim fat and seam fat are removed from the cut, weighed, and included in the reported refuse. Weights are determined for the whole retail cut as purchased, and for each component (e.g., separable lean, separable fat, refuse, etc.). Nutrient analyses are conducted on the separable lean and the separable fat. The external trim fat and the seam fat are combined for analyses and reported as separable fat. The nutrient values for separable lean and separable fat are weighted for their respective contributions to the whole retail cut and reported as "separable lean and fat". For cooked beef cuts, the cuts are cooked with the separable fat intact. Nutrient data for separable fat, separable lean only, and separable lean and fat of cooked cuts are analyzed or calculated as described above.

The analytical nutrient data include the mean nutrient value, the standard error given to three decimal places, and the number of observations on which the values are based. For many food items, mean values are given without an accompanying standard error and number of samples. These values are either calculated by pooling data by or by weighting means (e.g., All Grades), by applying cooking yields or nutrient retention factors, or by imputation from a different, closely related food. For raw beef items and unheated cured items, nutrient values are estimated on the known content of that nutrient in the lipid (fatty acids), total solids (cholesterol), moisture-free, fat-free solids (minerals), or protein (water-soluble vitamins) fraction of a similar food.

Nutrients

Nutrient information for SR can be found under "File Content" in the documentation. However, some nutrient information specific to beef products are included here. Nutrient data are obtained for moisture, protein, ash and total fat. The values for protein are calculated from the content of total nitrogen (N) in the food using the conversion factor recommended by Jones (Jones, D.B., 1941). The specific factor applied to beef items is 6.25. The carbohydrate content of uncured products (except some organ meats)

consisting entirely of beef is negligible. For such foods, the carbohydrate content is assigned a zero value. The sum of the percentages of water, protein, total lipid, and ash may not necessarily equal 100 percent for those foods showing zero carbohydrate because the amounts of each of these constituents are determined independently.

For heart, liver, kidney, tongue, and cured products (foods expected to contain carbohydrate), the carbohydrate value is calculated as the difference between 100 and the sum of the percentages of water, protein, total lipid, and ash. If the total of these constituents for any item is more than 100 due to analytical variation, the carbohydrate content is assigned a zero value.

Food energy is expressed in terms of both kilocalories and kilojoules. (One kilocalorie equals 4.184 kilojoules.) The data are for physiologic energy values remaining after losses due to digestion and metabolism have been deducted. Further discussions on energy and caloric factors used in SR can be found in the "Food Description File" of the general documentation.

The specific calorie factors used for calculating energy values in beef products are:

	Kcal/g
Protein	4.27
Fat	9.02
Carbohydrate	3.87

The carbohydrate factor of 3.87 is used for some organ meats and some cured products. The factor of 4.11 is used for tongue. The factors are based on the Atwater system for determining energy values. Details of the derivation of these factors are outlined in Agriculture Handbook No. 74 (Merrill, A.L. and Watt, B.K., 1973). Because the level of carbohydrate in separable lean and separable fat is insignificant, no carbohydrate factor is needed for most beef products.

Description of Projects

The studies documented in these notes on beef represent only data collected since 1998.

Selected cuts, 1/8 inch external trim fat.

A collaborative study was funded by the Beef Checkoff Program and conducted by USDA, America's Beef Producers, and Texas A&M University to determine the food and nutrient composition of 13 raw and cooked retail cuts for inclusion in the USDA National Nutrient Database for Standard Reference.

Sampling and fabrication. Carcasses (n=20) were selected from two packing plants, one in the Texas Panhandle and the other in Nebraska. Ten USDA Choice and ten USDA Select carcasses (yield grade 2 and 3) were selected for the study. These

carcasses represented the approximate distribution found in the US beef supply according to the National Quality Beef Audit – 1998 (Boleman, S.L. *et al*., 1998). All carcasses were shipped to Texas A&M University for fabrication of the following retail cuts: arm roast, bottom round roast, bottom round steak, brisket – flat half, eye of round roast, flank steak, round tip roast, small-end rib steak, tenderloin steak, tri-tip (bottom sirloin butt) roast (boneless and defatted), top loin steak, top round steak, and top sirloin steak. Cuts were assigned randomly to the following external fat trim levels: 0.0 cm (0 inch trim), 0.3 cm (1/8 inch trim), or 0.6 cm (1/4 inch trim). External fat was measured at five points, the points connected, and with a scalpel, the fat was removed half the thickness of the cut. This procedure was repeated on the other side, thus removing the excess fat completely. One additional steak was assigned to a raw treatment and trimmed to 0.3 cm. Three of the cuts (flank steak, round tip roast, and tri-tip roast) had no external fat and were therefore assigned to the 0.0 cm group for both preparations (raw and cooked). Dried surfaces, extending chine bones, minor muscles, and muscle pieces were trimmed from all cuts. All cuts were vacuum packed individually, labeled, and frozen at -23°C for further dissection and cooking. Additional details on fabrication have been previously published (Wahrmund-Wyle, J.L. *et al*., 2000).

Cooking procedures. (Wahrmund-Wyle, J.L. *et al*., 2000). Retail cuts destined for cooking were thawed overnight in a cooler at 5°C, weighed, and cooked as follows: arm roast, bottom round steak, and brisket were braised; bottom round roast, eye of round roast, round tip roast, and tri-tip roast were roasted; and flank steak, small-end rib steak, tenderloin steak, top loin steak, top round steak, and top sirloin steak were broiled.

For braising, cuts were browned for 4-8 min (time being size-dependent) in a preheated Farberware® Dutch Oven placed on top of a conventional range. After browning, the cuts were covered with 90-180ml distilled water, placed in a preheated conventional gas oven at 325°F (163°C) and simmered in a covered vessel to an internal temperature of 185°F (85°C).

Cuts for roasting were placed on wire racks with the fat side up, when possible, and cooked in a conventional gas oven (preheated to 325°F (163°C) to an internal temperature of 140°F (60°C). For broiling, cuts were cooked on electric Farberware® Open-Hearth Broilers (model 350A) to an internal temperature of 149°F (65°C). The internal temperature of each retail cut was monitored by inserting copper constantan thermocouples into the geometric center of the cut and recording the data on Honeywell recorders. After cooking, cuts were wrapped in plastic wrap and chilled (2-3°C) overnight (Jones, D.K. *et al*., 1992). Each cut was weighed prior to and after cooking for calculation of cooking yield.

Sample preparation. Individual samples from all cuts, both raw and cooked, were carefully dissected to separate and weigh the various cut components. These components included separable lean, external fat, seam fat, and waste such as bone and heavy (non-edible) connective tissue. The separable lean included muscle, intramuscular fat, and connective tissue that would be considered edible. External fat is the fat on the outside of the cut. Seam fat included intermuscular fat depots within the